

MySmartStudy

Empowering Teacher Trainees through Artificial Intelligence and
Collaborative Digital Learning

**Akmal Hakimi Bin Abd Rashid, Mohamed Azeem Bin Sulaiman,
Dr. Yusnizal Binti Ismail, Rohana Binti Bachik, Mohd Khairi Bin Sharifuddin, Marhaini Binti
A. Rahman**

UniKL MIIT & IPG Kampus Perempuan Melayu

Representing: IPG Kampus Perempuan Melayu

Survey Data Analysis & System Design Report | n = 103 Respondents

1. Background of Problem

First-year teacher trainees at IPG Kampus Perempuan Melayu Melaka face significant challenges in transitioning from the structured secondary school environment to the autonomous learning expected in tertiary education. To validate these challenges and inform the design of MySmartStudy, a comprehensive survey was conducted among 103 first-year students across 9 classes (PI 2, PI 3, BM 3, BM KBT, SEJ, PK 2, TESL 1, TESL 3, BM 1) using a 30-item online questionnaire.

1.1 Digital Readiness of Students

The survey revealed that 84% of respondents (87 out of 103) are comfortable or very comfortable with using digital technology for studying. Specifically, 34% are very comfortable and 50% are comfortable. Only 1% reported being slightly uncomfortable, while 15% remained neutral. This confirms strong digital readiness among the target users, validating that a digital learning platform would be well-received.

**Digital Technology Comfort Level
(n=103)**

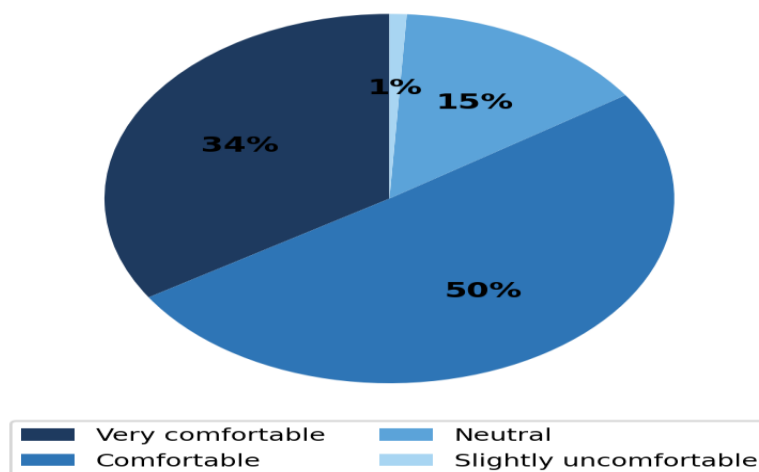


Figure 1: Digital Technology Comfort Level Among Respondents (n=103)

1.2 Poor Note Organisation and Memorisation Reliance

A critical finding is that 88% of students (91 out of 103) sometimes, often, or very often rely on memorisation because their notes are not well organised. Only 12% rarely resort to memorisation. The breakdown: 61% sometimes, 23% often, and 4% very often. When asked about their biggest frustrations, the top responses were: difficulty finding information quickly (28%), too many files or notebooks (14%), and notes scattered across platforms (13%).

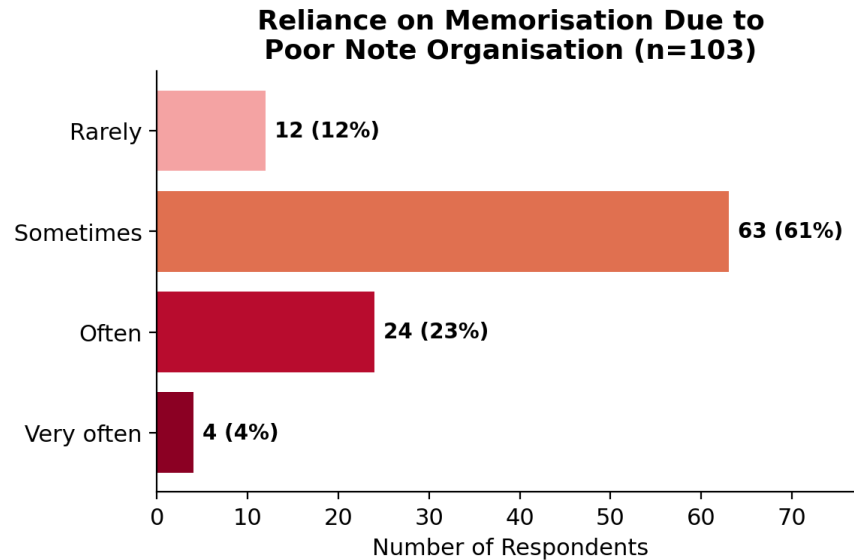


Figure 2: Reliance on Memorisation Due to Poor Note Organisation (n=103)

1.3 Thinking Map Usage Challenges

While 64% of students report being confident or very confident in using Thinking Maps, 74% have been unsure which template best suited their topic (68 sometimes + 8 frequently). The main barriers to more frequent usage are time constraints (30 respondents), difficulty choosing the right map (29), and a combination of both (11). Furthermore, 70% face handwriting or space issues when drawing complex maps manually (58 sometimes + 14 often), confirming the need for a digital mind map editor.

Confidence in Using Thinking Maps (n=103)

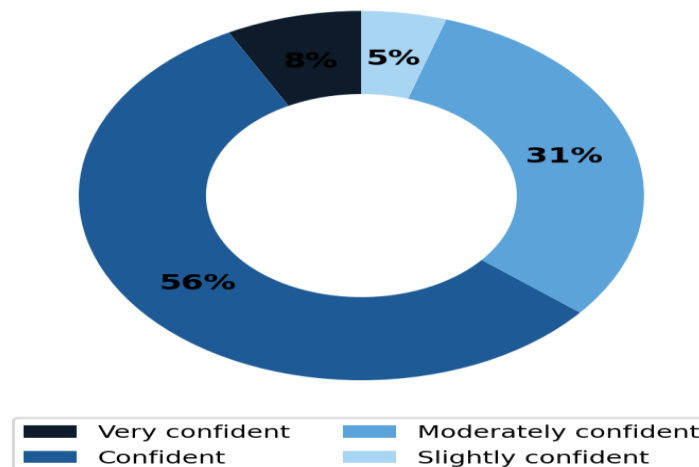


Figure 3: Confidence in Using Different Types of Thinking Maps (n=103)

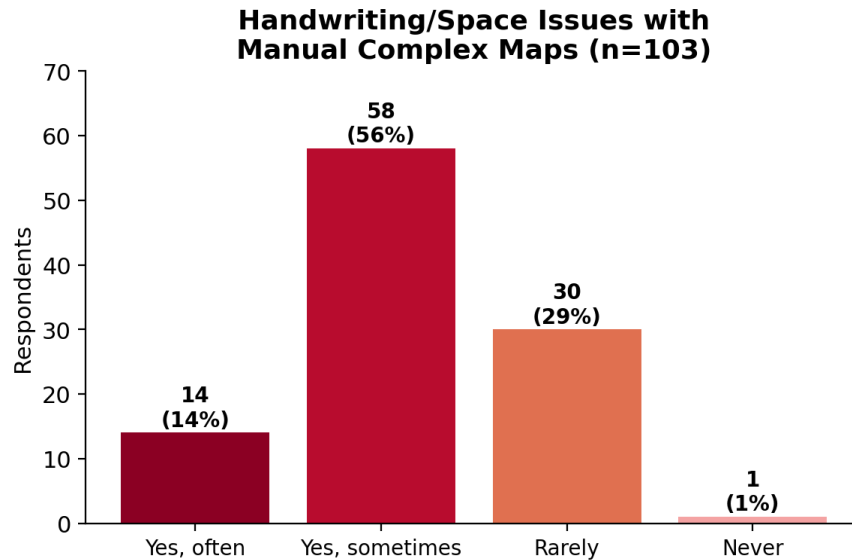


Figure 4: Handwriting and Space Issues with Manual Complex Maps (n=103)

1.4 Collaboration and Group Work Challenges

In group work, 82% of respondents have experienced duplicated work or version confusion (8 very often + 25 often + 51 sometimes). Encouragingly, 80% view real-time collaborative editing of Thinking Maps positively (65 positive + 17 very positive), and 90% find comment and feedback features on shared maps useful (61 useful + 32 very useful).

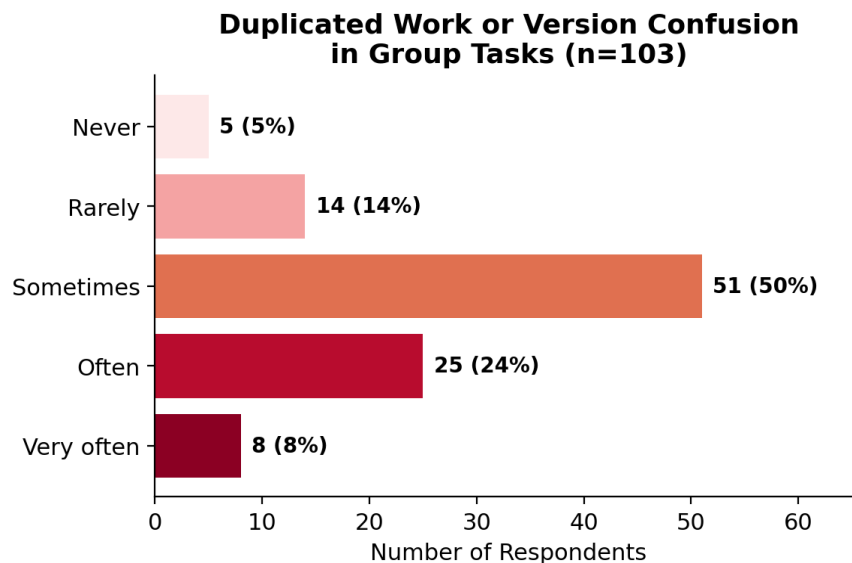


Figure 5: Duplicated Work or Version Confusion in Group Tasks (n=103)

2. Design of the Product

Based on the survey findings from 103 respondents, MySmartStudy was designed as a comprehensive multi-platform AI-enhanced Learning Management System. The system architecture and feature set were informed by the specific needs expressed by the students. This section presents the product highlights, system framework diagrams, and survey validation of key features.

2.1 Development Methodology

The Waterfall Software Development Life Cycle (SDLC) was selected due to the fixed academic scope and structured documentation requirements of the Final Year Project. The methodology consists of five sequential phases: Requirement Analysis (informed by the 103-respondent survey), System Design, Implementation, Testing (unit, integration, and UAT), and Deployment & Maintenance.

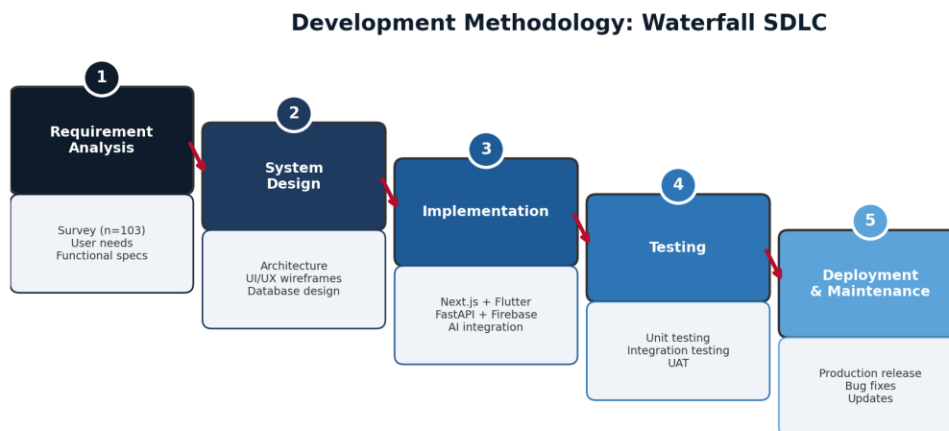


Figure 6: Waterfall SDLC Development Methodology

2.2 System Architecture Framework

MySmartStudy is built on a hybrid multi-tier architecture consisting of six interconnected layers: the User Layer (students, lecturers, admin), the Frontend Layer (Next.js 16 web app and Flutter mobile app), the API Gateway (FastAPI with 41 REST API routers), the Service Layer (authentication, courses, maps, quizzes, gradebook, analytics, AI, and messaging modules), the Data Store (Firebase Firestore with Firebase Authentication), and the AI Engine (Google Gemini 2.5 Flash with ChromaDB for RAG).

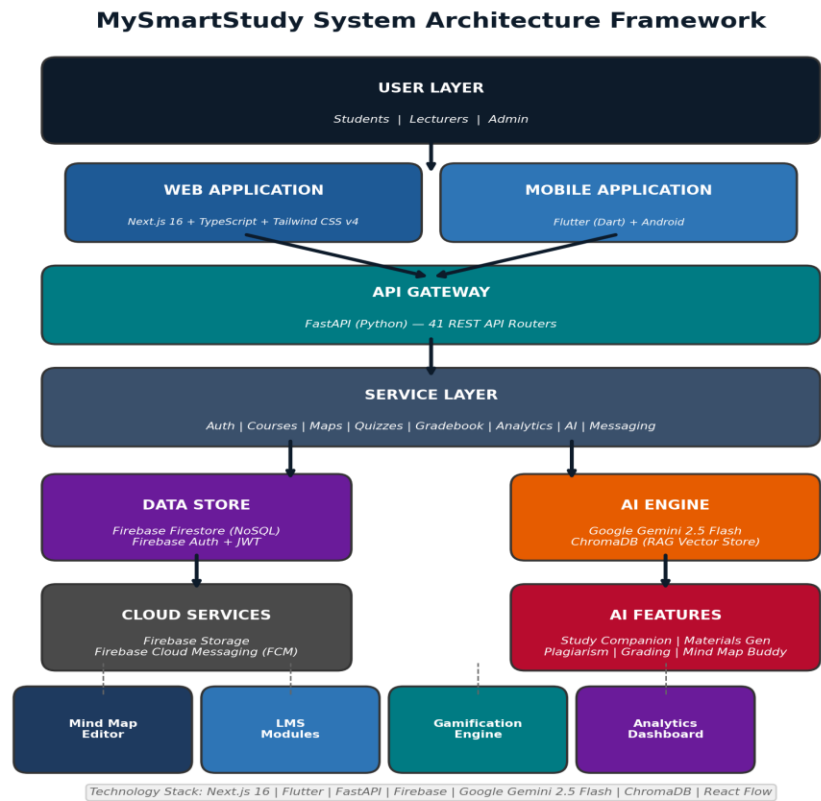


Figure 7: MySmartStudy System Architecture Framework

2.3 User Process Flow

The user process flowchart illustrates how students and lecturers interact with the system from registration through to continuous learning. After authentication via Firebase Auth + JWT, the system routes users based on their role (student or lecturer) to their respective dashboards. Students progress through course enrollment, material access, assignment submission, mind map creation, AI study tools, and gamification tracking. Lecturers manage courses, resources, assignments, reviews, AI-assisted evaluation, and analytics. Both roles interact through real-time collaboration features and share a common Firebase Firestore cloud sync layer.



Figure 8: MySmartStudy User Process Flow

2.4 AI-Powered Features Workflow

MySmartStudy integrates seven AI-powered features through a centralized AI Feature Router. When a user submits a request, the router directs it to the appropriate AI module: Study Companion (chat-based academic support), Study Materials (auto-generated summaries, flashcards, practice quizzes), Exam Planner (personalized study schedules), Mind Map Buddy (RAG-based node suggestions), Plagiarism Detection (similarity scoring for lecturers), Grading Assistant (AI-recommended grades and feedback), and Timetable Analysis (schedule conflict detection). All modules are powered by the Google Gemini 2.5 Flash AI engine with ChromaDB providing RAG capabilities for contextual responses.

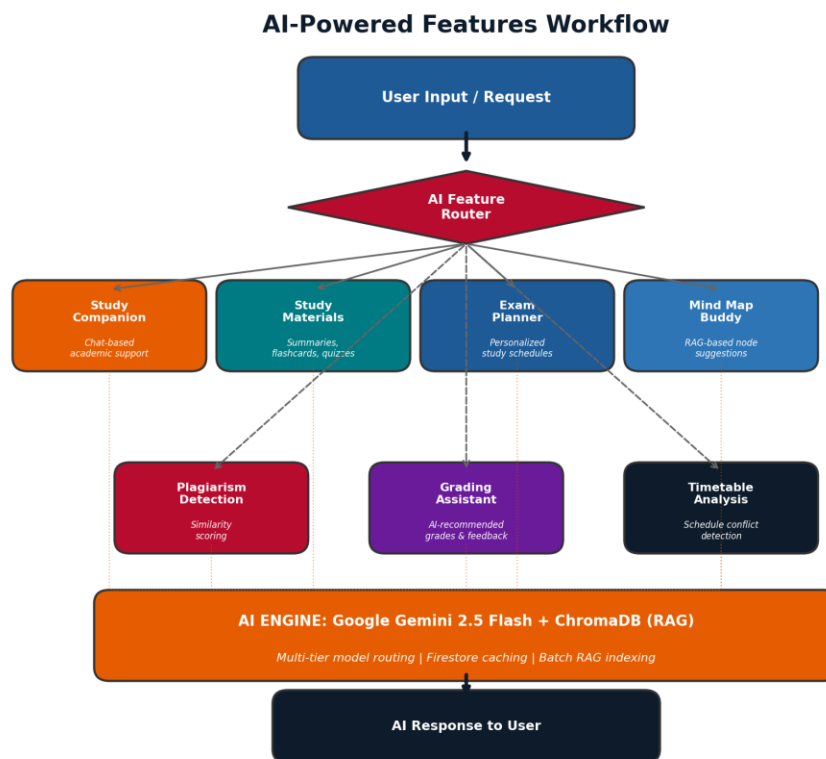


Figure 9: AI-Powered Features Workflow

2.5 Product Highlights

Key features designed in direct response to survey data:

- Collaborative Mind Map Editor with 15+ templates, real-time collaboration, presence indicators, and AI node suggestions via RAG — addressing the 74% who are unsure which template to use and 70% facing handwriting/space issues.
- AI Study Companion powered by Google Gemini 2.5 Flash — supporting the 88% who rely on memorisation.
- AI-Generated Study Materials (summaries, flashcards, practice quizzes) — addressing the need for better note organisation.

- Gamification Engine with 12 auto-awarded badges, login streaks, participation scoring — responding to the 80% who find gamification motivating.
- Complete LMS Suite (courses, assignments, quizzes, gradebook, attendance, forums, chat, messaging, calendar) — replacing fragmented tools.
- Lecturer Analytics Dashboard with at-risk student identification and submission trend monitoring.
- Progress Tracking Dashboard — responding to the 78% who say it would help manage studies.

2.6 Survey Validation of Key Features

The survey data strongly validates the feature design. 83% find automatic Thinking Map recommendation helpful or very helpful. 95% believe pre-filled templates would save time. 80% find gamification motivating. 78% confirm progress tracking would help.

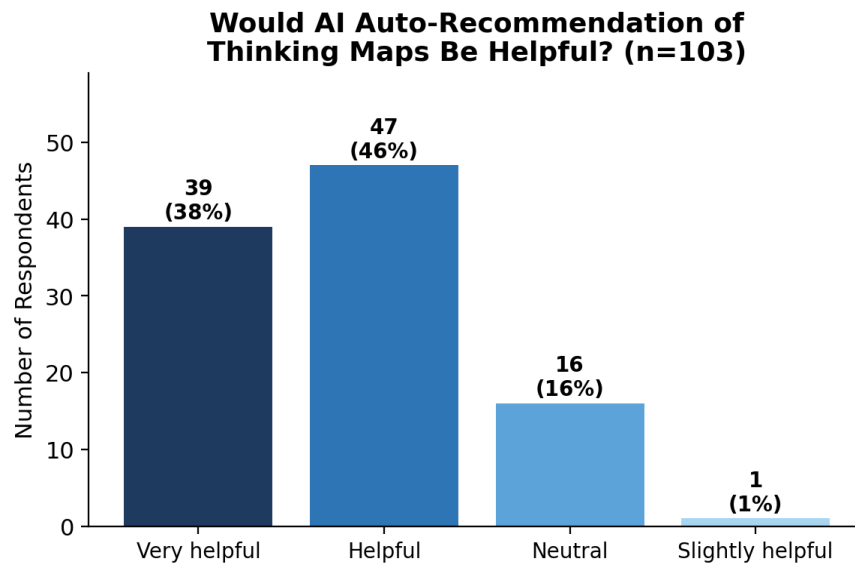


Figure 10: Demand for AI Auto-Recommendation of Thinking Maps (n=103)

Do Gamification Features (Badges, Streaks) Motivate You? (n=103)

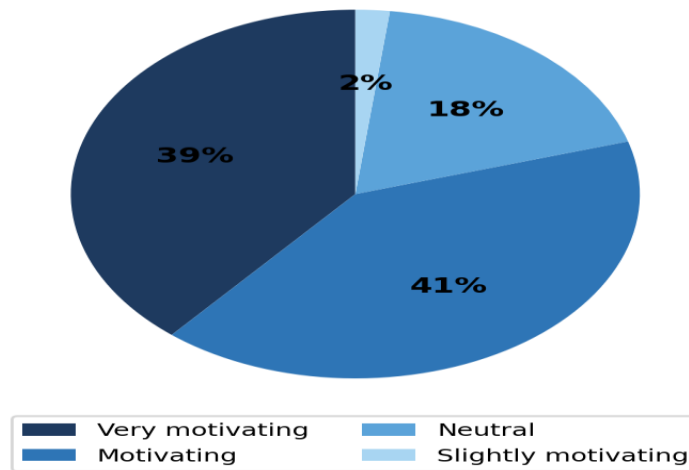


Figure 11: Motivation from Gamification Features (n=103)

2.7 Technology Stack Summary

Layer	Technology	Purpose
Frontend (Web)	Next.js 16, TypeScript, Tailwind CSS v4, React Flow	Full LMS dashboard with mind map editor
Frontend (Mobile)	Flutter (Dart)	On-the-go access, AI tools, notifications
Backend API	FastAPI (Python) — 41 routers	REST API for all system operations
Database	Firebase Firestore (NoSQL)	Real-time cloud data storage & sync
Authentication	Firebase Auth + JWT (HS256)	Secure role-based access control
AI Engine	Google Gemini 2.5 Flash	Study companion, materials, plagiarism
Vector Store	ChromaDB	RAG-based contextual AI responses
Mind Maps	React Flow (@xyflow/react)	Interactive collaborative visual editor
Notifications	Firebase Cloud Messaging	Real-time push notifications

3. Impact Towards Education Stakeholders

The survey results from 103 respondents demonstrate strong demand and readiness for MySmartStudy. The impact analysis below is supported by quantitative evidence.

3.1 Impact on Students

Survey Finding	Statistic	MySmartStudy Solution
Rely on memorisation due to poor notes	88% (91/103)	AI summaries, mind maps, organised notes
Unsure which Thinking Map to use	74% (76/103)	AI auto-recommendation engine
Handwriting/space issues with maps	70% (72/103)	Digital mind map editor, unlimited canvas
Version confusion in group work	82% (84/103)	Real-time collaborative editing
Find gamification motivating	80% (82/103)	12 badges, streaks, achievements
Want progress tracking dashboards	78% (80/103)	Student dashboard with tracking
Would pilot test & recommend app	89% (92/103)	High adoption potential confirmed

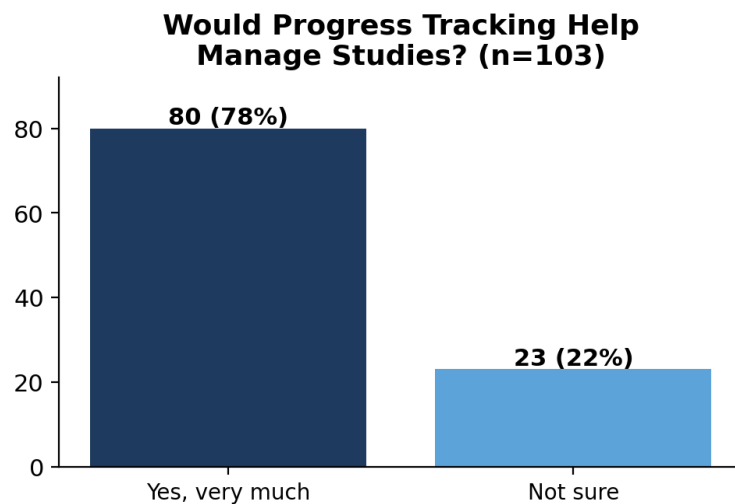


Figure 12: Student Demand for Progress Tracking Dashboards (n=103)

3.2 Impact on Lecturers

MySmartStudy provides lecturers with real-time analytics dashboards that track student engagement, submission trends, and course completion rates. The at-risk student identification feature enables early intervention before poor performance manifests in formal assessments. AI-powered plagiarism detection and grading assistance improve evaluation efficiency and consistency. The mind map review interface allows lecturers to assess student understanding through visual knowledge representations.

3.3 Impact on the Institution

With 84% of students already digitally comfortable and 89% willing to participate in pilot testing and recommend the platform, the institutional adoption potential is very high. The centralized platform replaces fragmented tools, standardizes digital learning practices, and produces future educators who are digitally literate. The audit logging system ensures accountability, while the admin dashboard provides oversight.

Willingness to Pilot Test & Recommend the App (n=103)

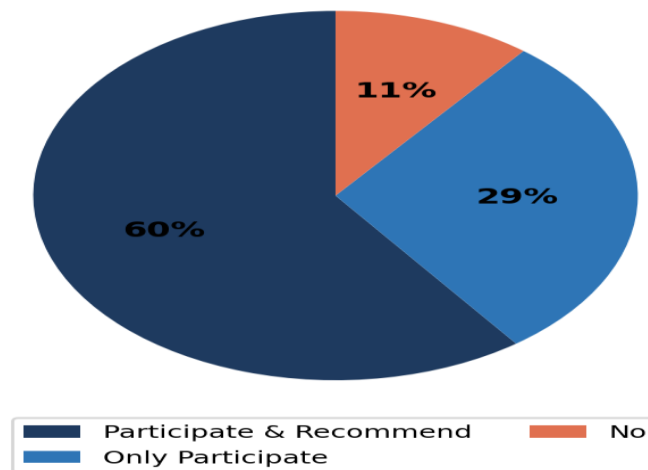


Figure 13: Willingness to Pilot Test and Recommend MySmartStudy (n=103)

3.4 Summary

The survey data from 103 respondents provides strong empirical validation for MySmartStudy. Key statistics confirm that the identified problems are real and widespread (88% memorisation reliance, 82% version confusion, 74% template uncertainty), the proposed solutions match user needs (83% want AI recommendations, 80% want gamification, 95% want pre-filled templates), and adoption readiness is high (84% digitally comfortable, 89% willing to pilot and recommend). MySmartStudy is a data-driven response to documented challenges in teacher training education.